



Fecal microbiota transplantation for Crohn's disease-like intestinal lesions arising after allogeneic stem cell transplantation

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Abstract

Several cases of inflammatory bowel disease (or similar gastrointestinal lesions) arising after allogeneic hematopoietic stem cell transplantation have been reported, but the effect of intestinal dysbiosis on development of these lesions remains unclear. We performed fecal microbiota transplantation (FMT) and 16S rRNA microbiome analysis in a patient who developed Crohn's disease-like lesions after allogeneic transplantation. A 62-year-old woman underwent haploidentical stem cell transplantation from her daughter to treat double-hit lymphoma relapsed after chimeric antigen receptor T-cell therapy, and achieved remission without developing acute graft-versus-host disease. Eight months later, she developed Crohn's disease-like intestinal lesions after cytomegalovirus enteritis. Her condition did not improve with the conventional treatment, so she underwent FMT from her daughter as part of a clinical trial. Diarrhea gradually improved, and follow-up endoscopy 4 months after the FMT showed ulcer healing and scarring. The 16S rRNA analysis revealed a reduction in the relative abundance of the *Enterococcus* genus after FMT, suggesting that dysbiosis may have contributed to lesion development. The patient is currently on a regular diet, with no symptom recurrence, and the primary disease remains in remission. Although this outcome suggests that FMT is effective, careful patient selection is required to reduce the risk of FMT-associated sepsis.

Keywords Fecal microbiota transplantation · Stem cell transplantation · Crohn's disease · 16S rRNA analysis · *Enterococcus*

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Introduction

Although allogeneic hematopoietic stem cell transplantation (allo-HSCT) is extensively performed as a curative treatment for various hematopoietic disorders, it is associated with various gastrointestinal complications, including gut graft-versus-host disease (GVHD), cytomegalovirus (CMV) enteritis, and intestinal transplant-associated thrombotic microangiopathy. Previous reports have suggested a therapeutic effect of fecal microbiota transplantation (FMT) in steroid-resistant gut GVHD [1, 2] and a potential prophylactic effect against GVHD onset [3]. Several cases of post-allo-HSCT have been reported to develop Crohn's disease (or similar gastrointestinal lesions) [4–8], as well as ulcerative colitis [9, 10]. There are many reports of successful treatment with anti-tumor necrosis factor-alpha antibodies [5, 6] and thalidomide [7]. Additionally, one patient with *Clostridioides difficile* enteritis complicated by Crohn's disease achieved remission of both conditions through a combination of FMT

and immunosuppressive therapy [8]. However, the previous studies have not used molecular biological techniques to reveal the relationship between disease activity and intestinal microbiota. Here, we present the results of a 16S rRNA analysis in a patient with Crohn's disease-like intestinal lesions that developed after allo-HSCT, who achieved remission following FMT alone.

Case presentation

A 62-year-old woman was referred to our hospital with a left rib tumor. A diagnosis of double-hit lymphoma was made based on histopathological examination. One month after receiving chimeric antigen receptor T-cell therapy (CAR-T, tisagenlecleucel) for chemotherapy-resistant disease, the patient experienced a relapse with multiple subcutaneous lesions. Three months later, she underwent human leukocyte antigen haploidentical hematopoietic stem cell transplantation in non-remission, with her first daughter as the donor. The patient was treated with 180 mg/m² of fludarabine, 140 mg/m² of melphalan, 2.5 mg/kg of rabbit anti-human thymocyte globulin, and 4 Gy of total body irradiation. Next, the patient received an infusion of peripheral blood stem cells with a CD34-positive cell count of $0.16 \times 10^6/\text{kg}$ and bone marrow stem cells with a total nucleated cell count of $2.2 \times 10^8/\text{kg}$. Steroids, tacrolimus, and short-term methotrexate were administered for GVHD prophylaxis. Five months after the allo-HSCT, the patient achieved complete remission without developing acute GVHD (Fig. 1). Seven months after transplantation, the patient developed persistent diarrhea and gastrointestinal bleeding, and an endoscopy revealed multiple punched-out ulcers. Polymerase chain reaction (PCR) and mucosal tissue pathology detected CMV, and a diagnosis of CMV enteritis was made (Figs. 1 and 2a). One month after foscarnet administration, mucosal PCR confirmed the disappearance of CMV. However, the presence of multiple longitudinal ulcers and a cobblestone appearance on endoscopy indicated a transition to Crohn's disease (Figs. 1 and 2b). Laboratory results are shown in Table 1. There were no noticeable perianal or upper gastrointestinal lesions at the diagnosis and during Crohn's disease-like colitis. Although mucosal histopathology did not reveal non-caseating epithelioid cell granulomas typical of Crohn's disease, a probable diagnosis of Crohn's disease was made based on the clinical presentation [11]. No evident recurrence of lymphoma was observed in positron emission/computed tomography; however, fluorodeoxyglucose uptake in the descending colon suggested intestinal inflammation (Fig. 1).

Steroid and immunosuppressive agents were not increased due to the risk of relapse of the primary disease. Mesenchymal stromal cells for acute GVHD were not used

due to the absence of typical pathological and endoscopic findings. Biologic drugs for Crohn's disease were not used due to concerns about the risk of recurrent infections, such as CMV. Three months after the diagnosis of Crohn's disease-like intestinal lesions, the patient was discharged after her symptoms stabilized with the administration of 5-aminosalicylate (5-ASA), fasting, and oral intake of elemental supplements (Fig. 1). However, there was no improvement in endoscopic mucosal findings. After 8 months of outpatient follow-up, the patient participated in a clinical trial of FMT (19 months after the allo-HSCT). This interventional study was approved by the Institutional Review Board of Osaka University Hospital (jRCT No. CRB5180007), and written informed consent was obtained from all participants. This clinical study included adult patients with refractory diarrhea and microscopic dysbiosis refractory to standard therapies, regardless of the primary disease. Healthy volunteers (18–70 years of age) or family donors without microscopic dysbiosis were included as potential donors and screened for pathogens in their stool samples. Those with predispositions to inflammatory bowel disease (IBD), irritable bowel syndrome, chronic diarrhea, severe constipation, malignancies under treatment, autoimmune diseases, severe obesity, Parkinson's disease, autism, depression, or Creutzfeldt–Jakob disease were excluded from potential donors. The patient's daughters and husband were screened as potential donors after obtaining written informed consent. The first daughter, a stem cell donor, was a carrier of *Escherichia coli* O1; therefore, the second daughter was selected as the FMT donor. After collecting the donor's whole stool, the microbial extracts were promptly suspended in saline under refrigerated conditions, gauze filtered, and stored at -80°C . The extracts were sprayed onto the patient's colon using a lower endoscope, without antimicrobial premedication. A 16S rRNA analysis was performed using donor and patient samples at multiple time points (before FMT, on the day of FMT, and +1, +3, +7, +14, and +28 days after FMT). The 16S rRNA analysis was performed as previously described [12]. Gram staining of the stool samples collected before and one day after FMT is shown in Fig. 2c, f [13].

After FMT, the diarrhea symptoms gradually improved, and 5-ASA agents were reduced and discontinued. Oral intake was eventually changed to a regular diet, and no recurrence of symptoms was observed (Fig. 1). The 16S rRNA analysis revealed short-term changes in gut microbiota composition following FMT, characterized by a decrease in the relative abundance of the *Enterococcus* genus (Fig. 2g). Twenty-eight days after the FMT, Simpson's diversity index recovered from 0.65 (on the day of the FMT) to 0.80. The resolution of ulcerative lesions and the presence of scarring were observed on follow-up lower endoscopy 4 months after FMT (Fig. 2d, e). Albumin levels improved, reflecting remission of enteritis

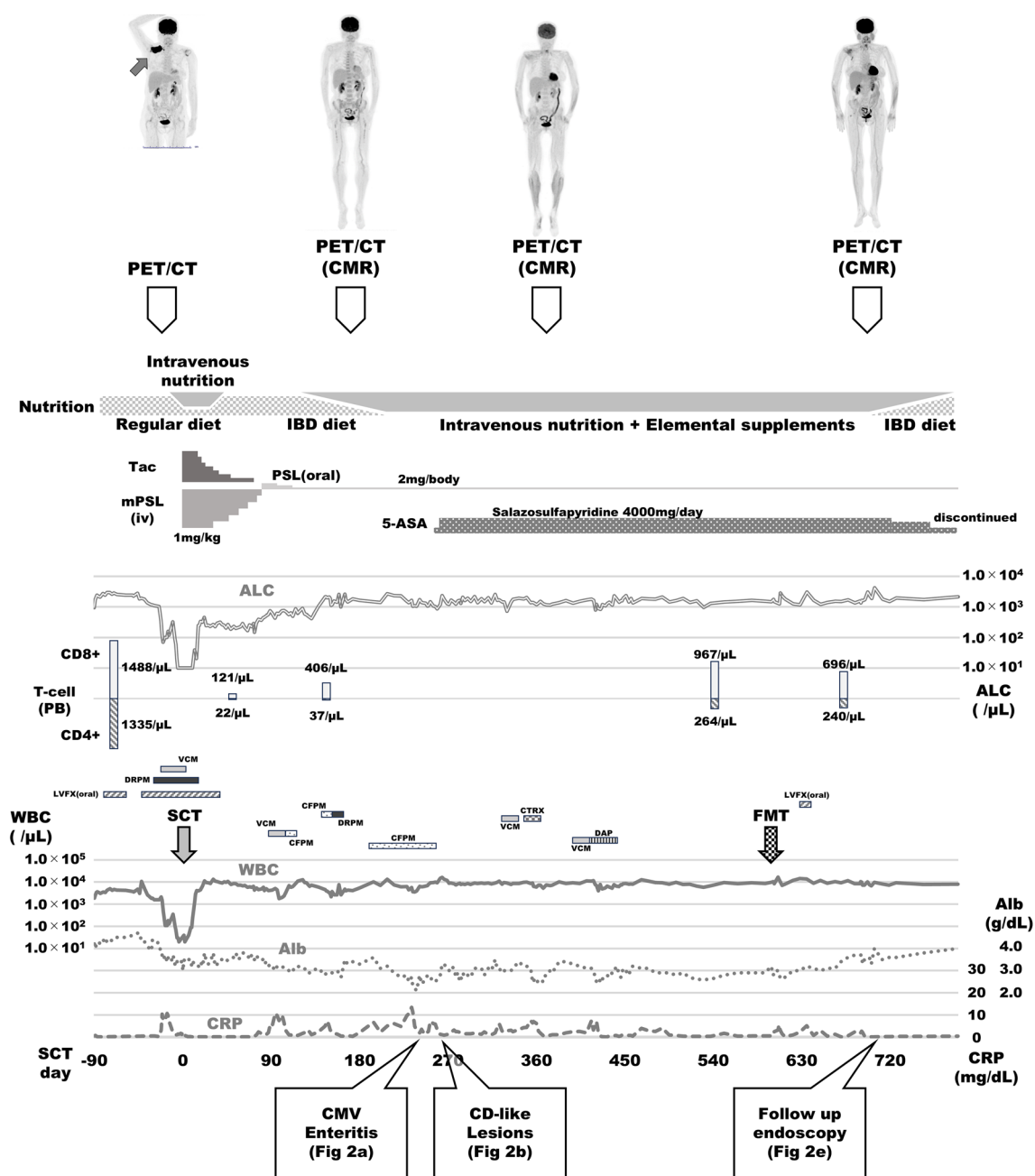
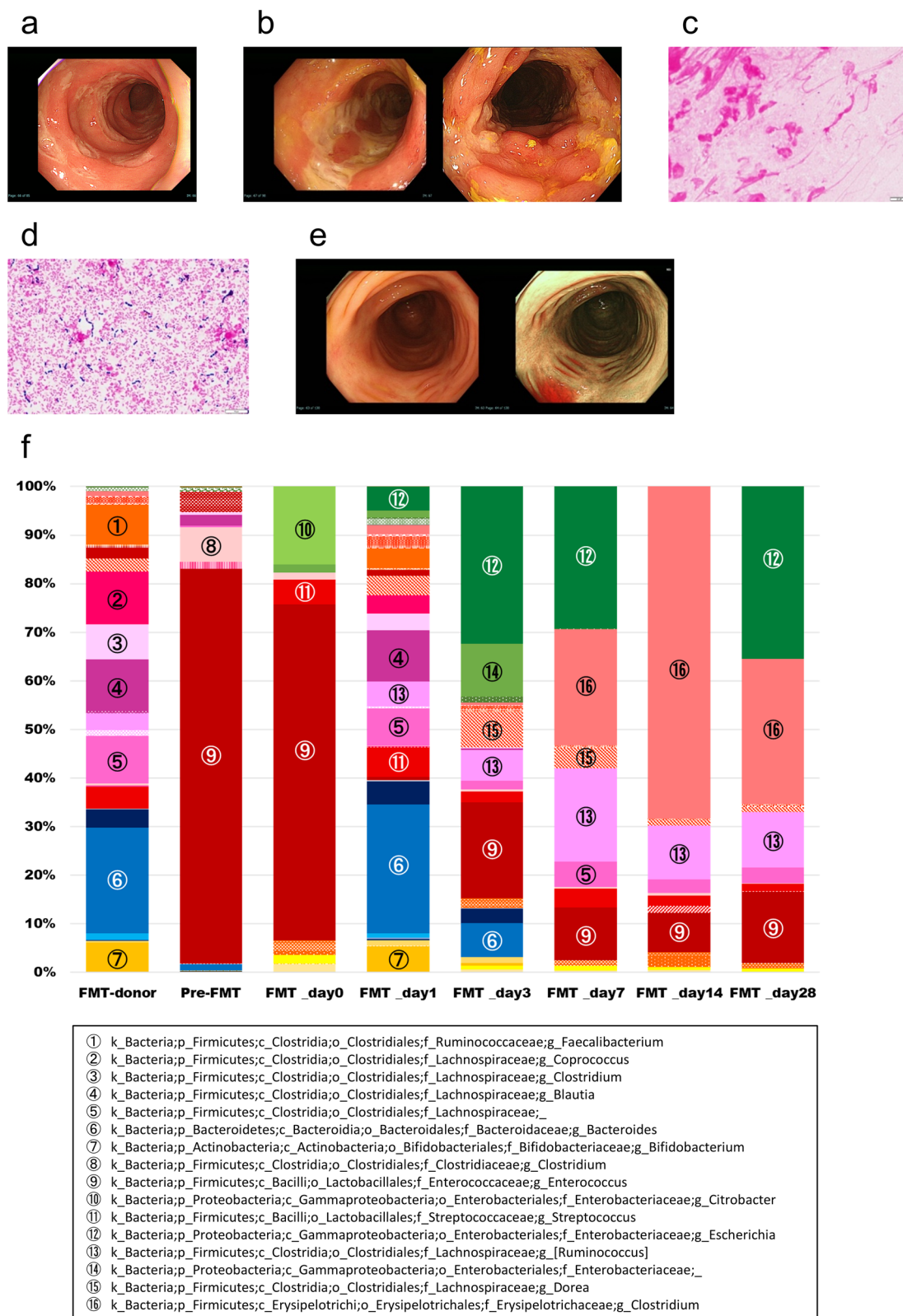


Fig. 1 Clinical course. *PET/CT* positron emission tomography/computed tomography, *CMR* complete metabolic response, *IBD* inflammatory bowel disease, *Tac* tacrolimus, *(m)PSL* (methyl) prednisolone, *iv* intravenous, *ASA* aminosalicylic acid, *PB* peripheral blood, *ALC* Absolute lymphocyte count, *VCM* vancomycin, *DRPM* doripenem,

LVFX levofloxacin, *CFPM* cefepime, *CTR* ceftriaxone, *DAP* daptomycin, *SCT* stem cell transplantation, *FMT* fecal microbiota transplantation, *CMV* cytomegalovirus, *CD* Crohn's disease, *WBC* white blood cell, *CRP* C-reactive protein, *Alb* albumin

(Fig. 1). Three months after FMT (22 months after the allo-HSCT), the primary disease remained in complete remission (Fig. 1). After CMV colitis, the patient had been taking 900 mg/week of valganciclovir as maintenance

therapy for CMV colitis, which was discontinued 8 months after the FMT (27 months after the allo-HSCT); however,



no relapse of CMV infection was observed. At 3 years

Fig. 2 Endoscopic findings, Gram staining of the stool samples, and results of 16S rRNA analysis. **a** Endoscopic findings 7 months after allogeneic transplantation; multiple punched ulcers were observed, consistent with CMV enteritis. **b** After 1 month, follow-up endoscopy showed alterations of Crohn's disease-like intestinal lesions. **c** Gram staining of the stool sample (8 months post-transplant) was consistent with the timing of (**b**), and only a few Gram-positive cocci were seen, suggesting severe dysbiosis. **d** Gram staining of the stool sample (1 month after FMT) showed Gram-positive cocci, bacilli, and Gram-negative bacilli, with diversity restored. **e** Follow-up endoscopy 3 months after FMT (normal and narrow-band images); the ulcer surfaces healed, leaving only scars. **f** Results of 16S rRNA analysis: changes in the bacterial flora were observed within a few days after FMT, along with a decrease in the relative abundance of the *Enterococcus* genus over time. Circled numbers were assigned to the major genera. The yellow lineage represents the *Actinobacteria* phylum, the blue lineage represents *Bacteroidetes* phylum, the red lineage represents the *Firmicutes* phylum, and the green lineage represents the *Proteobacteria* phylum. *k* kingdom, *p* phylum, *c* class, *o* order, *f* family, *g* genus

post-allo-HSCT, the patient remains alive without recurrence of the primary disease or GVHD.

Discussion

In this study, we present a case of Crohn's disease-like gastrointestinal lesions that developed after allo-HSCT and were successfully treated with FMT. The patient was initially diagnosed with CMV enteritis and treated with antiviral drugs. Follow-up endoscopic findings led to a probable diagnosis of Crohn's disease, based on established diagnostic criteria [11]. Because the involvement of CMV enteritis could not be completely ruled out in this patient's gastrointestinal lesions, we described these lesions as "Crohn's disease-like" to emphasize the clinical picture. However, the actual T lymphocyte counts showed a recovery trend immediately just before FMT, yet there was no improvement in the intestinal lesions. In addition, a dramatic improvement in the enteritis was observed after FMT, although the T-cell count did not change significantly. Based on these findings and the clinical manifestations, we believe that there is minimal likelihood of a direct involvement of viral infections in this lesion. In this patient, CMV infection preceded the development of Crohn's disease-like intestinal lesions. Cytokine production associated with CMV infection contributes to the exacerbation of IBD [14]. In this patient, CMV infection may be involved in the development of Crohn's disease-like intestinal lesions. In recent years, possible interactions between CMV reactivation, the gut microbiota, and the immune system have been reported [15, 16]. FMT has been reported in pediatric patients with ulcerative colitis complicated by CMV enteritis, suggesting that FMT could be a safe and effective treatment option [17]. In this patient, prophylactic CMV antivirals were discontinued later. Alterations

in the gut microbiota due to FMT may have affected the risk of CMV reactivation.

It is presumed that the use of broad-spectrum antimicrobials around the time of the allo-HSCT contributed to the establishment of an *Enterococcus*-dominant intestinal microbiota. The 16S rRNA analysis confirmed that the *Enterococcus*-dominant gut microbiota was altered by FMT. This suggested that gut dysbiosis was involved in the development of gastrointestinal lesions. Several studies reported that the pathogenicity of *Enterococcus* may contribute to the etiology of IBD [18, 19] and that *Enterococcus* spp. may be involved in various complications following allo-HSCT. We previously reported a correlation between an increased relative abundance of *Enterococcus* after allo-HSCT and poor prognosis [12]. An association has been reported between the abundance of *Enterococcus*, an increase in GVHD, and higher mortality rates in patients and mouse models after allogeneic transplantation [20]. Potential new strategies for preventing acute GVHD using phage-derived antibacterial enzymes directed at pathogenic *Enterococci* spp. have also been reported [21].

Several clinical trials of FMT have recently been conducted in patients with IBD to modify gut microbiota and improve disease activity [22]. FMT is expected to be effective in the field of hematology [1–3]. However, recipients of allo-HSCT during the acute phase are generally at a high risk of bloodstream and gastrointestinal infections, due to neutropenia, mucosal fragility, and the use of immunosuppressive drugs or steroids. The development of bloodstream infections originating from FMT donors has been reported [23]. However, another study suggests that the pathogens causing bloodstream infections after FMT originate from the recipient's flora [24]. Therefore, the safety of FMT in allo-HSCT recipients requires further investigation. We performed FMT in this patient for intestinal lesions that developed over 6 months post-transplantation. Given that the patient had no complications of enteric origin before FMT (including bloodstream infections) and required minimal antimicrobial therapy, we considered it safer to perform FMT in this patient than in those in the acute phase after hematopoietic cell transplantation. Furthermore, the minimal administration of antimicrobials around the time of the FMT may have contributed to the recovery of the intestinal microbiota following the FMT.

Conclusion

We safely performed FMT in a patient with Crohn's disease-like gastrointestinal lesions that developed following allo-HSCT. Although FMT is expected to be effective in treating such lesions by modifying the intestinal

Table 1 Laboratory tests at the diagnosis of Crohn's disease-like intestinal lesions

Complete blood cell count			Biochemistry tests			Infection tests	
WBC	9240	/μL	TP	5.6	g/dL	CMV-DNA	Not detected
Mye	1	%	Alb	2.7	g/dL	EBV-DNA	Not detected
St	4	%	AST	18	U/L		
Seg	66	%	ALT	29	U/L	PCR(<i>Mycobacterium</i>)	
Lym	14	%	ALP	80	U/L	<i>Avium</i>	Not detected
Mo	13	%	γGTP	28	U/L	<i>Intracellularae</i>	Not detected
Eo	2	%	LDH	152	U/L	<i>Tuberculosis</i>	Not detected
Ba	0	%	T-Bil	0.3	mg/dL		
RBC	311 × 10 ⁴	/μL	BUN	17	mg/dL	IGRA	
Hb	9.1	g/dL	Cre	0.64	mg/dL	<i>Tuberculosis</i>	(−)
Plt	7.4 × 10 ⁴	/μL	Na	135	mEq/L		
			K	4.8	mEq/L	Microscopic test	
Coagulation tests			Cl	105	mEq/L	Protozoa	(−)
PT-INR	1.07		Ca	9.4	mg/dL		
APTT	30.0	sec	CRP	7.12	mg/dL	Culture test	
(normal range: 24–39 s)						(intestinal fluid collected by endoscopy)	
Fibrinogen	663	mg/dL				<i>Enterococcus sp.</i>	3+
D-dimer	1.63	μg/mL				CNS	1+
						<i>Mycobacterium</i>	(−)
						CD toxin test	(−)

WBC white blood cell, Mye myelocyte, St band neutrophil (stab cell), Seg segmented neutrophil, Lym lymphocyte, Mo monocyte, Eo eosinophil, Ba basophil, RBC red blood cell, Hb hemoglobin, Plt platelet, PT-INR prothrombin time–international normalized ratio, APTT activated partial thromboplastin time, TP total protein, Alb albumin, AST aspartate aminotransferase, ALT alanine aminotransferase, ALP alkaline phosphatase, γGTP gamma-glutamyl transferase, LDH lactate dehydrogenase, T-Bil total bilirubin, BUN blood urea nitrogen, Cre creatinine, Na sodium, K potassium, Cl chloride, Ca calcium, CRP C-reactive protein, CMV cytomegalovirus, EBV Epstein–Barr virus, DNA deoxyribonucleic acid, PCR polymerase chain reaction, IGRA interferon-gamma release assay, sp. species, CNS coagulase-negative *Staphylococci*, CD *Clostridioides difficile*

microbiota, its use should be carefully considered, given the potential risk of infection from donor stool.

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Author contributions All authors contributed to the conception and design of the study. SK, RK, and KF treated the patient and drafted the manuscript. KS and TY planned and conducted the FMT study. DM and SN performed 16S rRNA analysis. MK performed pathological evaluations of the tissue. KN, AH, HK, TU, and JF treated the patient and discussed the etiology. NH, TT, and JO advised on the study and revised the manuscript. All authors revised and approved the final manuscript.

Data availability The datasets analyzed in this study were submitted to DDBJ/EMBL/GenBank databases under accession number PRJDB20424.

Declarations

Conflicts of interest The authors declare that they have no conflicts of interest.

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